

ROLE B PLAYBOOK

Page + charts · the photographer's screen

Pair project · Nathalia & Dafna · Role B

This is not a weather app. Weather is the input. The product is a decision tool for outdoor photographers in Israel.

Photographer asks: I have a portrait in Tel Aviv tomorrow at 17:00. Should I shoot then — and is there a better window?

The product formula

Weather Data → Analysis → Photography Score → Recommendation → Decision

Fill the # TODO (STUDENT) comments. Do not copy a finished weather app. One owner per file.

Your job in one sentence

You show a decision, not a weather dump. After Plan shoot the photographer must see: Photography Score, best shooting window, one calm sentence, then charts. You never parse JSON in app.py — you only display what Role A returns.

Your files — what each one is for

File	Function (what you write)	Notes
photo_planner/charts.py	photography_score_chart — score vs time (the product chart). temperature_chart · clouds_and_rain_chart.	No API calls. Return Plotly figures only.
app.py	Coordinator. Inputs are already KEEP. After the button: call client + scoring, then layout.	Catch WeatherError with st.error. No CSS. No extra pages.
.streamlit/config.toml	Warm theme already set. You may tweak later. Not required to learn the API.	Leave it for the MVP.
README.md	How a classmate runs the project: install, .env, streamlit run app.py, never commit the key.	Last polish before GitHub.

Your functions

Function	Receives	Must return
photography_score_chart	list of (HourlyConditions, score)	Plotly figure, score 0–100 vs clock time
temperature_chart	same list	Plotly figure of temperature_c

clouds_and_rain_chart	same list	Plotly figure of clouds and/or rain (pop × 100 if you want %)
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Page mockup — build this, nothing extra

One Streamlit page. Decision first, weather second. Skip: custom CSS, maps, login, comparing 3 cities, minute-perfect windows.

Order	What the photographer sees	Streamlit widget
0 · already done	Location, date, 17:00, photography type, Plan shoot	st.columns(4) + st.button (KEEP)
1	Tel Aviv · date · portrait	st.caption
2	87/100 · 18:00–21:00 · booked 17:00 score 83	st.columns(3) + st.metric
3	One or two recommendation sentences	st.success(...)
4	Photography Score during the day (mark the peak)	st.plotly_chart — full width
5	Temperature clouds & rain	st.columns(2) + two charts
6 optional	Why this hour? wind / rain / clouds	st.expander
Error	One red sentence, no traceback	except WeatherError: st.error(...)

TOGETHER — you both do these (do not split)

These are the only times you must sit together. The rest of the week you can work in parallel.

When	What you do together	Done when
Meeting 1 · start	Decide Role A / Role B (10 min). Copy <code>.env.example</code> → <code>.env</code> , paste the API key. Install with <code>pip install -r requirements.txt</code> .	Each person knows her files. <code>.env</code> exists. Never commit it.
Meeting 2 · notebook	Open <code>notebooks/01_forecast_api_colab.ipynb</code> . GET <code>/data/2.5/forecast</code> for Tel Aviv. Print <code>status_code</code> , then temp, wind, clouds, pop, sunset.	You see 200. Do not open Streamlit yet.
Meeting 3 · weights	On paper: for portrait, sunset, landscape write rain / wind / temperature / clouds weights that add to 1.0. Ask: what would cancel this shoot first?	Both can explain portrait vs sunset. Put numbers in <code>SHOOT_WEIGHTS</code> .
Meeting 4 · plug in	Connect <code>app.py</code> to real <code>get_forecast</code> + <code>score_forecast</code> . Try Tel Aviv portrait tomorrow 17:00. Try an error case.	The page shows a Photography Score. Mistakes show <code>st.error</code> , not a traceback.
Meeting 5 · ship	GitHub without secrets (<code>docs/DEPLOY.md</code>). Streamlit Cloud, secret name <code>OPENWEATHER_API_KEY</code> . Screenshots of score + error.	Public URL works. <code>.env</code> is not on GitHub.

You will NOT wait all week

Field names and function names are already in the starter files. Role A fills how they work. Role B can already use the names. The only way to block each other is changing a field name in `models.py` without saying so.

Your solo path (Role A is not a blocker)

Use fake scores until Meeting 4

Build two fake `HourlyConditions` (copy the field names from `models.py`) and pair them with scores 72 and 94. Draw the three charts and the metric layout with that list. When Role A finishes, replace the fake list with `score_forecast(forecast, shoot_type, date)`. Do not wait for the live API to start `charts.py`.

Order — one step at a time

1. Notebook with Role A until 200. Note photographer fields: time, temp, wind, clouds, pop, sunset.
2. Wait only for field names (already KEEP in `models.py`).
3. Meeting 3: weights on paper (you help decide; Role A codes them).
4. `charts.py` — score chart first, then the other two.
5. `app.py` layout with fake data matching the mockup.
6. Meeting 4: wire `get_forecast` + `score_forecast`, catch `WeatherError`.
7. Test locally, then README, then help deploy.

What you read from Role A (do not parse JSON)

hour.time_text · hour.temperature_c · hour.wind_speed · hour.cloud_cover · hour.rain_probability
scored_hours is a list of (hour, float). window is (start, end, best_score).

If you need a new field (for example visibility), Role A adds it on HourlyConditions first.

Your checklist

□ Notebook: photographer fields
Time, temperature, wind, clouds, rain chance, sunset.

□ Photography Score chart
Fill photography_score_chart. X = clock, Y = 0–100. No API in charts.py.

□ Decision row after Plan shoot
Three st.metric then st.success. Catch WeatherError with st.error.

□ Score chart on the page
Full-width st.plotly_chart. Mark the best hour if you can.

□ Temperature + clouds/rain
st.columns(2) and the other two chart functions.

□ Try the app locally
streamlit run app.py. Portrait Tel Aviv 17:00, sunset Haifa, missing-key error.

□ README for a classmate
Install, .env, streamlit run app.py, never commit the key.

□ Optional: mark preferred time
Highlight 17:00 on the score chart. Skip until the main chart works.

Tone of the page

Words for a photographer: "excellent", "windy for portrait", "prefer 18:00". Not "payload keys" or "status 200".
Default Plotly line charts — not 3D gauges. Theme is already in config.toml.

Rules: no secrets on GitHub, no copied weather app, no extra screens, one owner per file, weights decided together, fill TODOs yourself.